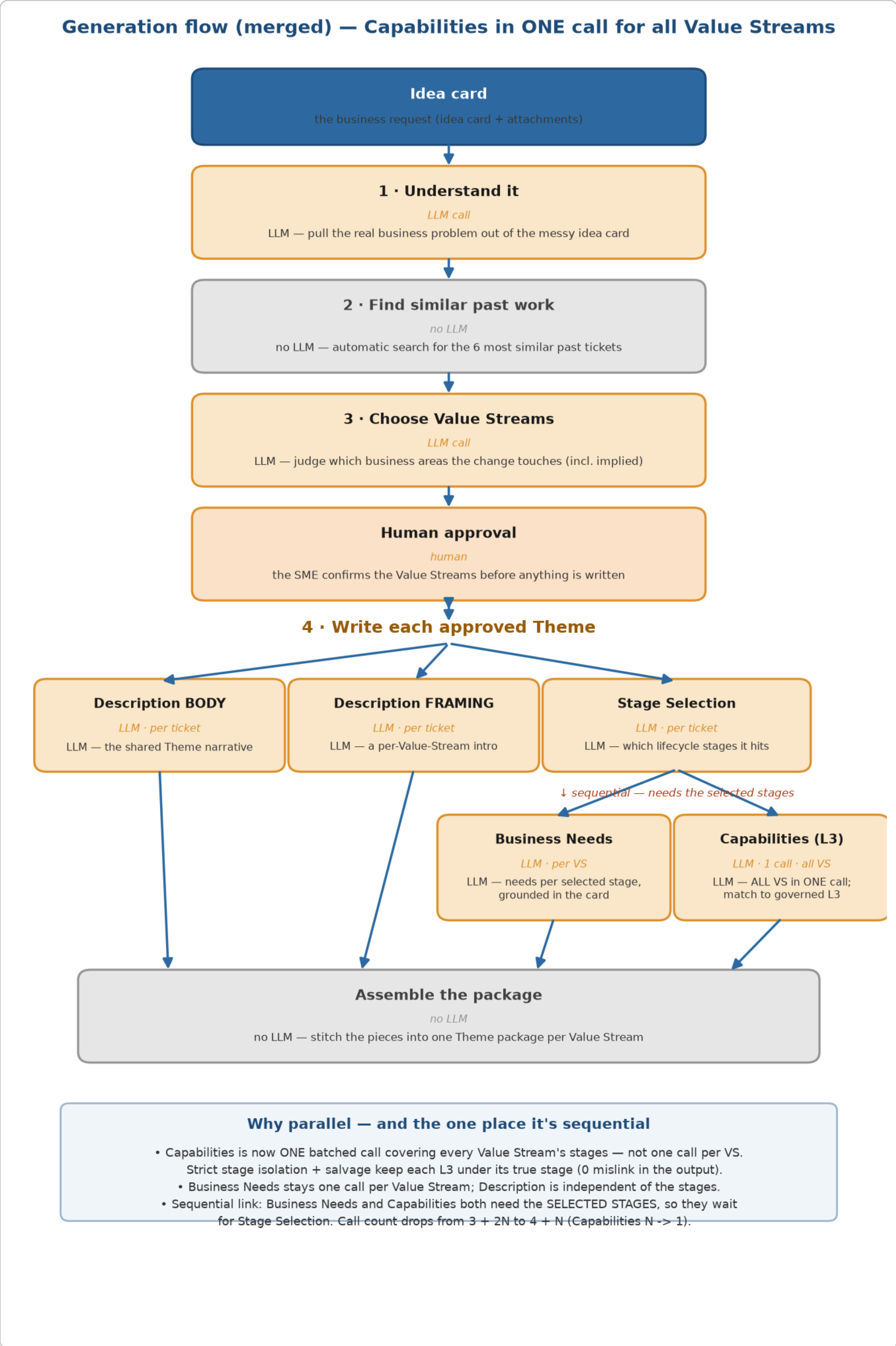
Generation flow — batching optimisations: cost & impact analysis



GPT-5-mini pricing throughout: **$0.25 / 1M input, $2.00 / 1M output.**

Baseline cost — current architecture (all per-VS), 10 Value Streams

Average tokens (the measured sample's idea cards sit below the 24k cap):

call	runs	input	output	cost
Condense	1	6,000	500	$0.0025
Choose Value Streams	1	12,000	1,500	$0.0060
Stage Selection	1	7,602	1,273	$0.0044
Description BODY + FRAMING	2	10,304	1,706	$0.0060
Business Needs	10	55,200	15,670	$0.0451
Capabilities (L3)	10	58,470	6,990	$0.0286
Total (average)	**25 calls**			**$0.093**

24k worst case — every call carries the full 24k raw idea-card text:

call	runs	input	output	cost
Condense	1	24,000	500	$0.0070
Choose Value Streams	1	30,000	1,500	$0.0105
Stage Selection	1	26,600	1,273	$0.0092
Description BODY + FRAMING	2	48,400	1,706	$0.0155
Business Needs	10	245,200	15,670	$0.0926
Capabilities (L3)	10	248,500	6,990	$0.0761
Total (24k)	**25 calls**			**$0.211**

So the baseline is **$0.093/ticket average, $0.211 worst case** (10 VS).

The changes and how they impacted quality

Merged Capabilities (L3) — all Value Streams in one call

metric	per-VS	merged
precision	0.53	0.43
recall	0.90	0.84
F1	0.67	0.57
mislink in output (after salvage)	0	0
hallucinated ids	0%	0%

Recall −6, strict precision −10 (much of which is plausible picks the ground truth didn't tag, not real error). Nothing hallucinated; mislink is 0 in the output either way. **Quality holds up well** — L3 is short *selection*, which batches cleanly.

Batched Business Needs — a few Value Streams per call

metric	per-VS	batched
faithfulness	0.895	0.715
hallucination	0.105	0.285
coverage	0.810	0.519

Faithfulness −18, coverage −29, hallucination +18. **Quality degrades materially.** Business Needs is long-form *writing*: the batched call emits **~920 output tokens per VS vs 1,567 per-VS** — each document is ~40% shorter, so it reflects fewer source facts and grounds fewer claims. The shorter output is the direct cause.

Cost comparison 1 — merged Capabilities (whole flow, 10 VS)

	baseline (per-VS)	merged Capabilities
LLM calls	25	**16**
cost — average	$0.093	**$0.082**
cost — 24k worst case	$0.211	**$0.157**

Capabilities collapses from 10 calls to 1: **9 fewer calls**, **−12% cost average**, **−26% at 24k** (the raw text is sent once instead of 10 times — the saving grows with idea-card size).

Cost comparison 2 — batched Business Needs (whole flow, 10 VS)

	baseline (per-VS)	batched Business Needs
LLM calls	25	**20**
cost — average	$0.093	**$0.073**
cost — 24k worst case	$0.211	**$0.168**

Business Needs goes from 10 calls to 5 (chunked): **5 fewer calls**, **−21% cost average**, **−20% at 24k**.

Verdict

change	calls saved	cost saved (avg / 24k)	quality
merged Capabilities	9	12% / 26%	small, acceptable trade
batched Business Needs	5	21% / 20%	**materially worse** (shorter, less-grounded docs)

Adopt merged Capabilities — it's the bigger call-count saving, the cost drop grows on large idea cards, and the quality trade is small (and nothing is hallucinated). **Keep Business Needs per-VS** — the saving is real, but it comes by making a prescriptive, architect-facing artifact ~40% shorter and notably less grounded, which is a bad trade.